INTRODUCTION

An oscilloscope displays how a voltage changes with time and is an essential instrument in electronics testing. Commercial oscilloscopes are expensive and bulky for students and hobbyists, so this project presents a **Pocket Digital Oscilloscope**, a low-cost and portable alternative built around the **ESP32** microcontroller, which captures an input signal and shows its waveform on a colour TFT display.

Advantages: The oscilloscope is low in cost, compact enough to fit in a pocket and displays the waveform in real time. It offers selectable input range and AC/DC coupling, works through simple push buttons without a computer, and uses easily available components that make it easy to build and repair.

Applications: It can be used as a learning tool in electronics laboratories, for testing and debugging hobby and DIY circuits, and for viewing low-frequency and audio-range signals and sensor outputs.

CIRCUIT DIAGRAM

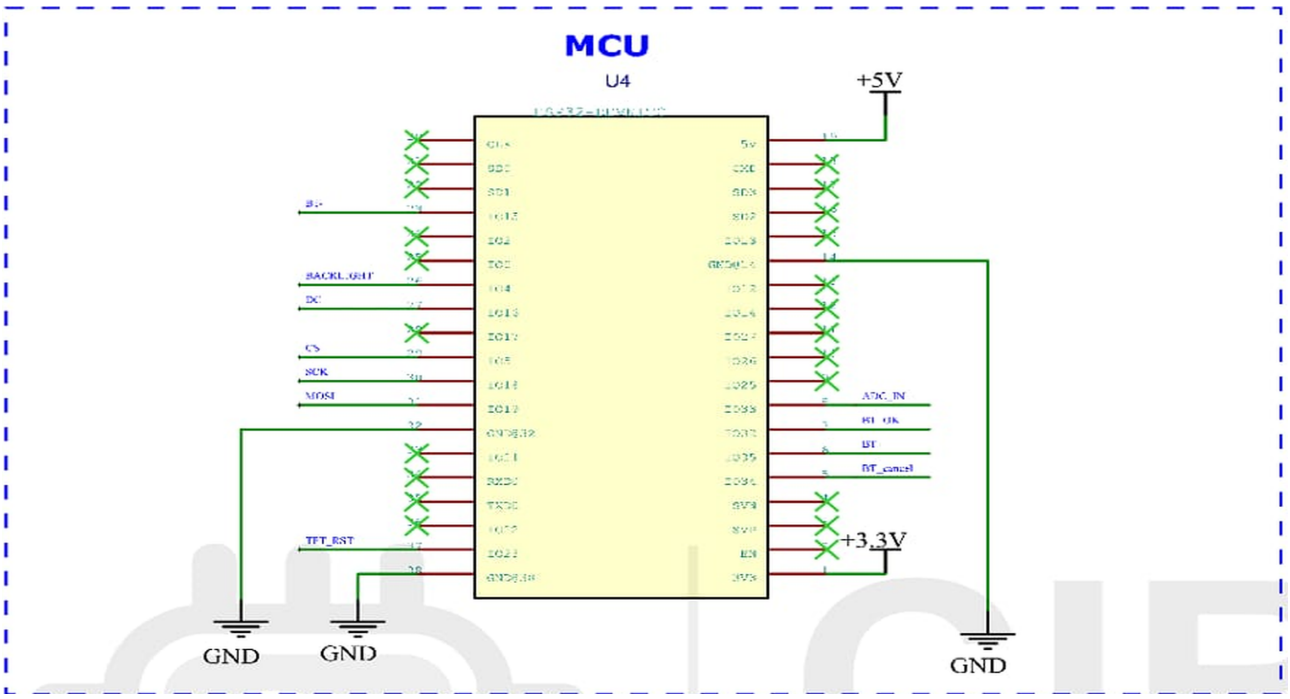


Fig. 1 MCU section (ESP32)

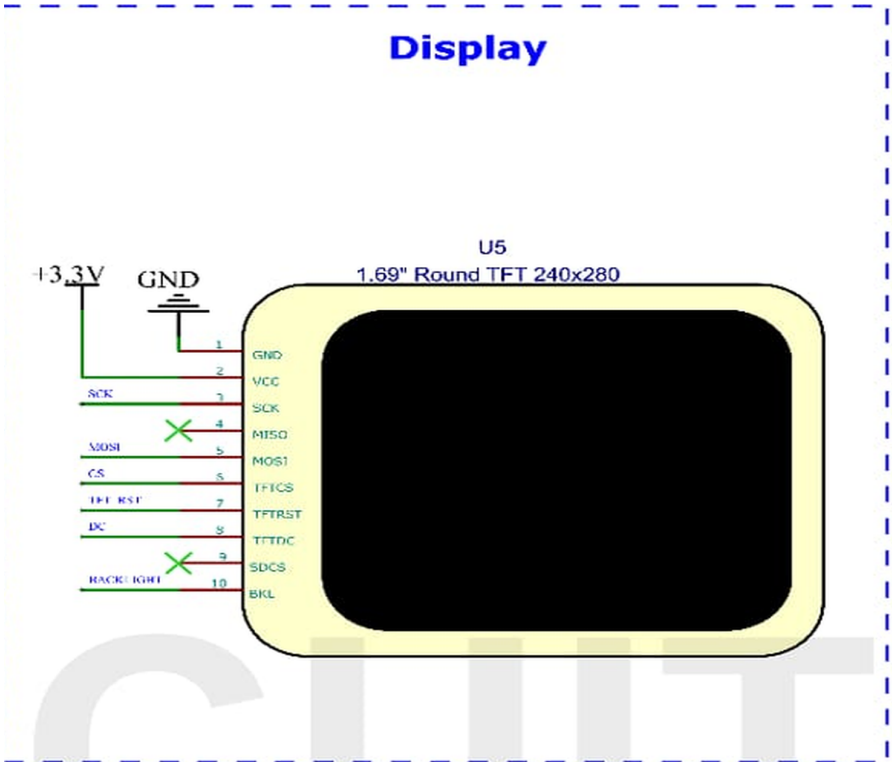


Fig. 2 Display section (1.69" TFT)

CIRCUIT DIAGRAM (Contd.)

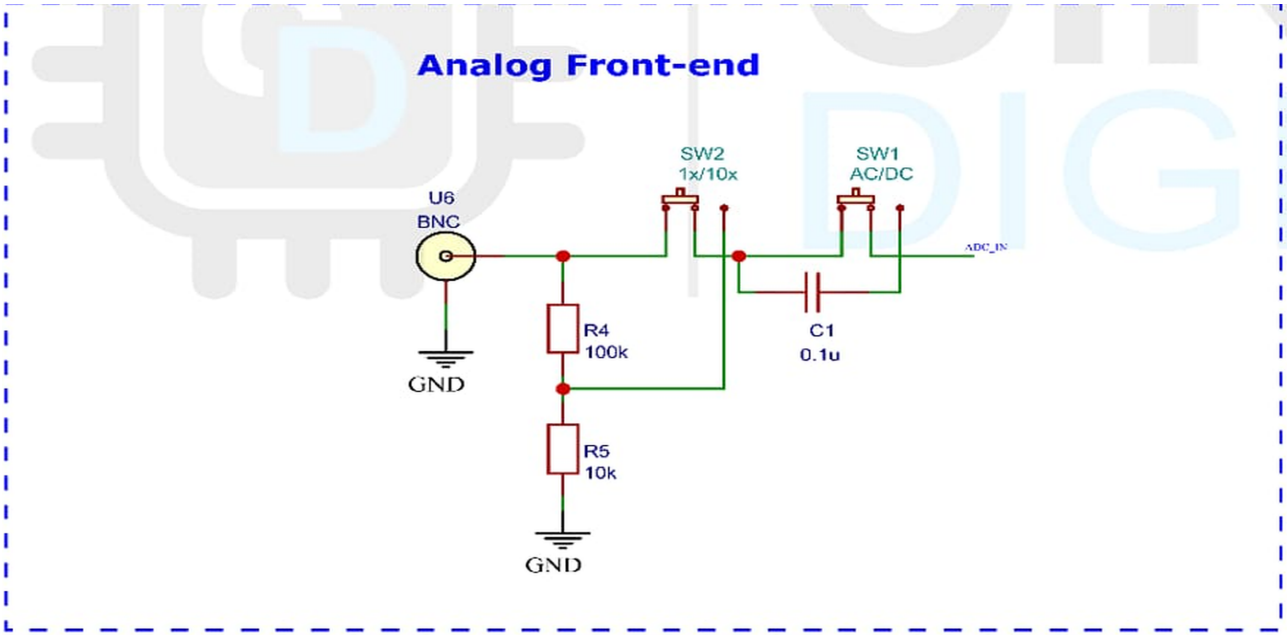


Fig. 3 Analog front-end section

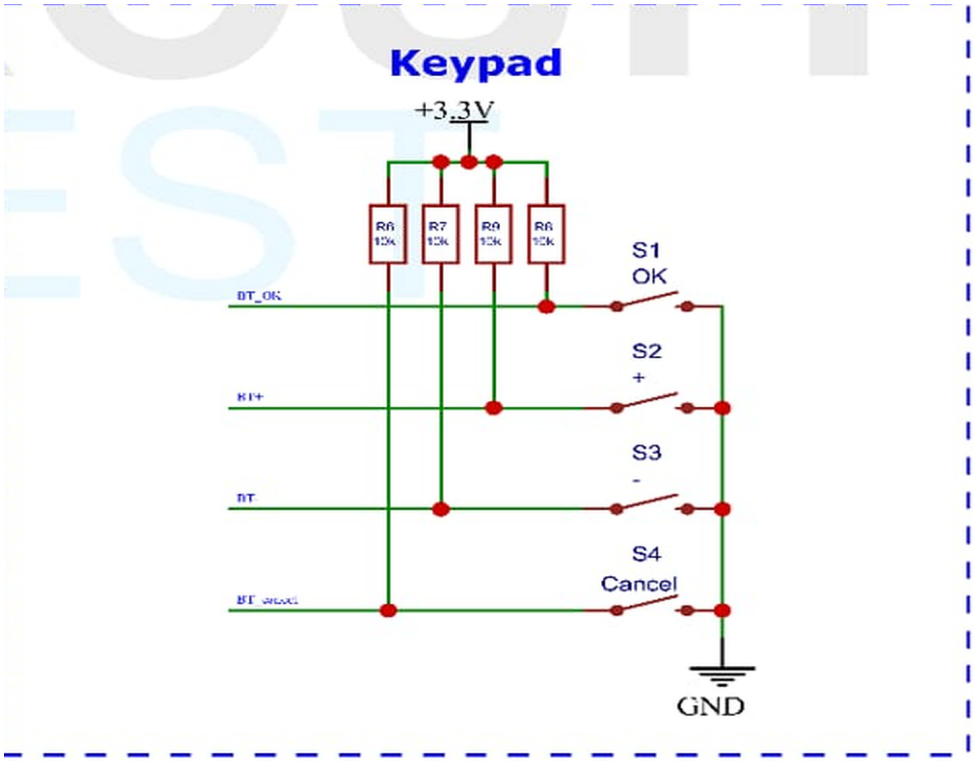


Fig. 4 Keypad section

LIST OF COMPONENTS

Sl. No.	Component	Specification	Qty.	Purpose
1	ESP32 development board	38-pin variant recommended	1	Main controller (ADC sampling, processing, display control)
2	TFT display (U5)	1.69″, 240×280, SPI	1	Signal visualisation
3	Tactile switches (S1–S4)	Standard 6 mm	4	User interface control (OK, +, −, Cancel)
4	SPDT switches (SW1, SW2)	Toggle type	2	Range (1x/10x) and coupling (AC/DC) selection
5	Resistor (R4)	100 kΩ	1	Input potential divider
6	Resistors (R5–R9)	10 kΩ	5	Potential divider (R5) and keypad pull-ups (R6–R9)
7	Capacitor (C1)	100 nF (0.1 µF) ceramic	1	AC coupling
8	BNC connector (U6)	Standard	1	Signal probe input
9	Signal probe	Standard	1	Measuring the signal
10	DC-DC buck converter	LM2596 module	1	Regulated power supply

PROTOTYPE OF THE POCKET DIGITAL OSCILLOSCOPE



Fig. 5 Prototype of the Pocket Digital Oscilloscope

The prototype is assembled in a compact enclosure. The TFT display is fitted in a cut-out on the front panel for viewing the waveform, and the four push buttons beside it form the keypad used to operate the instrument.

WORKING

The signal to be measured is connected through the signal probe to the BNC connector. The 1x/10x switch selects either the direct signal or a divided signal through resistors R4 (100 kΩ) and R5 (10 kΩ), which reduces large voltages to about one-eleventh. The AC/DC switch then passes the signal either directly or through capacitor C1 (0.1 µF), which blocks the DC component. This conditioned signal keeps the voltage within the safe input range of the ESP32 and is fed to its ADC input, where it is sampled at 50 kHz using the ADC and I2S interface. The ESP32 processes the samples in software, scales them according to the selected range and sends the waveform to the 1.69″ TFT display over SPI through the SCK, MOSI, CS, DC and RESET lines, so the waveform is shown in real time. Four push buttons (OK, +, − and Cancel) are connected to ESP32 input pins with 10 kΩ pull-up resistors to 3.3 V; pressing a button pulls its pin low, which the program uses to change settings and to confirm or cancel a selection. The circuit is powered from a regulated 5 V supply, obtained using the LM2596 buck converter, while the display and keypad run from 3.3 V.

RESULT

The Pocket Digital Oscilloscope was successfully designed and built using the ESP32 microcontroller, a TFT display and a simple analog front-end. The circuit captures the input signal through the probe and BNC connector, samples it at 50 kHz and displays the waveform on the TFT screen in real time. The 1x/10x range switch and the AC/DC coupling switch allow different kinds of signals to be viewed safely, and the push-button keypad allows the settings to be changed without using a computer.

The instrument works as a compact and affordable test tool and is suitable for viewing low-frequency and audio-range signals in educational and hobby applications.

CONCLUSION

The Pocket Digital Oscilloscope was designed as a compact and affordable instrument that brings basic oscilloscope functions to students and electronics hobbyists. Using the ESP32's built-in ADC and I2S interface, the circuit samples the input signal at 50 kHz, while the analog front-end keeps the input within the safe limits of the microcontroller. The captured waveform is displayed in real time on the TFT screen, and the push-button keypad lets the user operate the instrument without a computer.

The project shows that a useful test instrument can be built from a handful of inexpensive components, and it offers a good learning experience in analog signal conditioning, ADC sampling, SPI display interfacing and embedded programming. Because of its 50 kHz sampling rate, it is best suited to low-frequency and audio-range signals. The design can be improved in future by adding triggering, frequency measurement, waveform storage on an SD card and a faster ADC for higher-frequency signals.